

SOC Detection & Network Security Analysis Lab

Tools: Windows Event Viewer, Splunk Enterprise, Wireshark, Windows Defender Firewall, Nmap

Objective

Build an end-to-end SOC workflow from Windows authentication telemetry to SIEM detection and alerting, then validate supporting network-security concepts through packet capture, firewall control, and controlled scan traffic.

Lab Scope

- Generated and reviewed Windows authentication events including Event IDs 4624, 4625, and 4648.
- Distinguished local and network logons using logon type and source network address.
- Ingested Windows Security logs into Splunk and correlated failed and successful logons.
- Built and saved a scheduled detection for repeated failed logins followed by successful authentication.
- Compared HTTP with HTTPS/TLS in Wireshark and observed TCP and TLS handshakes.
- Validated Windows Firewall behavior by blocking outbound HTTP while allowing HTTPS.
- Generated controlled Nmap port-scan traffic and reviewed the pattern relevant to IDS/IPS detection.

SIEM Investigation and Detection

A controlled authentication sequence against the local test account soclab was used to create repeated failures followed by a successful login. Splunk reduced the event set and correlated activity by account, source address, logon type, and time.

Detection concept: 3+ failed logons followed by at least 1 successful logon from the same source within a 5-minute search window.

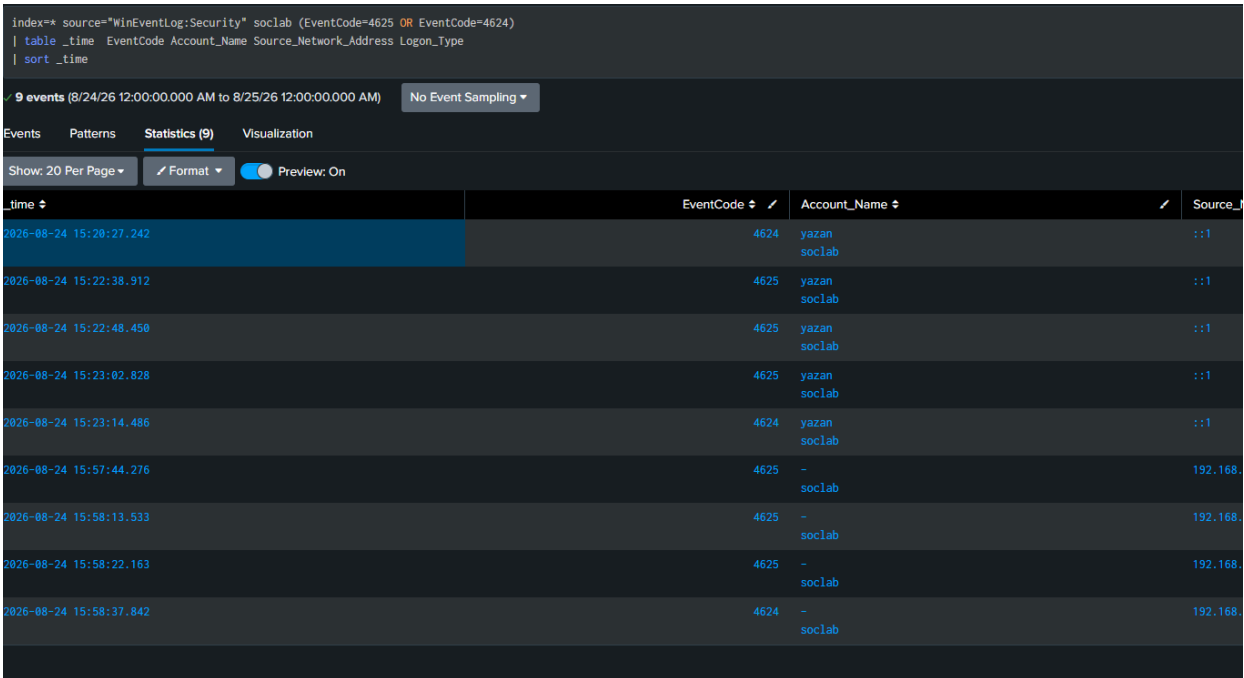


Figure 1. Splunk timeline showing failed logons followed by successful authentication.

HTTP, TLS and Firewall Validation

Wireshark showed readable HTTP request data such as URI, host, and user-agent, while HTTPS traffic was protected by TLS. The capture also showed the TCP three-way handshake, TLS Client Hello/Server Hello, and encrypted application data.

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1. Frame 1868: Packet, 588 bytes on wire (4704 bits), 588 bytes captured (4704 bits) on interface \Device\NPF_{0F762ABE-B0FA-4A5E-B2A1-CABA054E215}, id 0
  Ethernet II, Src: ASUSN1000G_1f:81:08 (10:7c:81:1f:81:08), Dst: Arcadyan_00:15:7c (c8:99:02:00:15:7c)
  Internet Protocol Version 4, Src: 192.168.1.166, Dst: 34.223.124.45
  Transmission Control Protocol, Src Port: 53884, Dst Port: 80, Seq: 1, Ack: 1, Len: 534
  Hypertext Transfer Protocol
    GET /online HTTP/1.1\r\n
      Request Method: GET
      Request URI: /online
      Request Version: HTTP/1.1
      Host: wonderfulidfunrainbow.neverssl.com\r\n
      Connection: keep-alive\r\n
      Cache-Control: max-age=0\r\n
      Upgrade-Insecure-Requests: 1\r\n
      User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/751.0.0.0 Safari/537.36\r\n
      Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7\r\n
      Referer: http://neverssl.com/\r\n
      Accept-Encoding: gzip, deflate\r\n
      Accept-Language: en-US,en;q=0.9,ar-AE;q=0.8,ar;q=0.7\r\n
      \r\n
    [Response in frame 1863]
    [Full request URI: http://wonderfulidfunrainbow.neverssl.com/online]
```

Figure 2. Readable HTTP request fields captured before TLS encryption.

A Windows Defender Firewall rule blocked outbound TCP port 80 while HTTPS on TCP 443 remained functional, confirming port-based traffic control.

IDS/IPS Fundamentals

A controlled Nmap TCP scan against the local gateway produced the characteristic pattern of one source probing many destination ports on the same target within a short period. This was used to reinforce the distinction between IDS detection/alerting and IPS detection with active blocking capability.

Outcome

The lab connected endpoint telemetry, SIEM detection, network traffic analysis, and basic security controls into one workflow. The main outcome was understanding how raw security events are collected, correlated, converted into detection logic, and surfaced to an analyst for investigation.